



LABORATORY JOURNAL (MP408)

Experimental Techniques – II Computational Methods in Physics

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Certificate

The Experiments Entered in This Journal Have Been Satisfactorily
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Computational Methods in Physics (MP408)

GNU Octave

PRACTICAL – 4

Interference of Two Identical 2-D Waves

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PROGRAM – 6

Aim

To write a GNU Octave program which describes the **interference of two identical 2-D spherical waves** from two coherent point sources (Young's double-slit geometry), and to obtain the static intensity fringe pattern on a screen for **seven different** sets of input parameters: wavelength λ , slit separation d , and screen distance D . The output is also saved as a 300 DPI PNG file.

Theory

Huygens' Principle (1678) states that every point on a wavefront acts as a secondary source of spherical wavelets. When two such sources are coherent (same frequency, fixed phase), their superposition produces a stationary interference pattern.

Consider two point sources (slits) at positions $(0, +d/2)$ and $(0, -d/2)$, with a screen at perpendicular distance D . For a point (X, Y) on the screen, the exact 3D distances to each slit are:

$$r_1 = \sqrt{X^2 + \left(Y - \frac{d}{2}\right)^2 + D^2}, \quad r_2 = \sqrt{X^2 + \left(Y + \frac{d}{2}\right)^2 + D^2} \quad (1)$$

The time-averaged intensity from superposition of two equal-amplitude waves is:

$$I(X, Y) = \cos^2\left(\frac{\pi |r_1 - r_2|}{\lambda}\right) \quad (2)$$

Fringe conditions:

- Bright fringes (constructive): $|r_1 - r_2| = n\lambda \Rightarrow Y_n \approx n\lambda D/d$
- Dark fringes (destructive): $|r_1 - r_2| = (n + \frac{1}{2})\lambda$

Fringe width:

$$\beta = \frac{\lambda D}{d} \quad (3)$$

Anti-aliasing (dynamic zoom): to prevent pixel aliasing when β is small, the screen half-width is set to $\min(15 \text{ mm}, 20\beta)$, ensuring at least ~ 20 fringes are resolved at adequate pixel density.

Set	λ (mm)	d (mm)	D (mm)	β (mm)	Key feature
1	5.0×10^{-4}	0.10	1000	5.00	Baseline
2	7.0×10^{-4}	0.10	1000	7.00	Wider ($\uparrow \lambda$)
3	5.0×10^{-4}	0.20	1000	2.50	Narrower ($\uparrow d$)
4	5.0×10^{-4}	0.10	2000	10.0	Wider ($\uparrow D$)
5	4.0×10^{-4}	0.05	500	4.00	Custom mix
6	5.89×10^{-4}	5.00	3000	0.354	Na D-line; very dense
7	2.0×10^{-4}	2.00	200	0.020	UV; extreme density

Program

The complete program is in P4.m (63 lines). Shown in full below:

Listing 1: 2D Wave Interference — Full Program (P4.m)

```

1 % Interference of two identical 2-D waves
2 % Static fringe pattern for 7 sets of input parameters
3
4 % Define 7 sets of input parameters: [lambda, d, D] (All units in mm
  )
5 params = [
6     0.0005,    0.10, 1000; % Set 1: Baseline
7     0.0007,    0.10, 1000; % Set 2: Larger wavelength
8     0.0005,    0.20, 1000; % Set 3: Larger slit gap
9     0.0005,    0.10, 2000; % Set 4: Larger screen distance
10    0.0004,    0.05,  500; % Set 5: Custom mix
11    0.000589,  5.00, 3000; % Set 6: 589nm, d=5mm, D=3m
12    0.0002,    2.00,  200 % Set 7: 200nm, d=2mm, D=0.2m
13 ];
14
15 % Create the figure window but keep it hidden while drawing to
    prevent GUI lag/jitter
16 fig = figure('Position', [50, 50, 1400, 800], 'Name', '2D Wave
    Interference', 'Visible', 'off');
17
18 % Set a balanced resolution
19 grid_size = 800;
20
21 for i = 1:7
22     lambda = params(i, 1);
23     d = params(i, 2);
24     D = params(i, 3);
25

```

```

26      % Calculate theoretical fringe width to prevent spatial aliasing
          (visual jitter)
27      fringe_width = (lambda * D) / d;
28
29      % Dynamically zoom the screen to show a clear pattern without
          pixel distortion
30      % Capped at +/- 15 mm max so the standard sets look normal
31      screen_max = min(15, 20 * fringe_width);
32
33      % Generate the grid for this specific subplot
34      [X, Y] = meshgrid(linspace(-screen_max, screen_max, grid_size),
          ...
35                          linspace(-screen_max, screen_max, grid_size));
36
37      % Calculate geometric distances
38      r1 = sqrt(X.^2 + (Y - d/2).^2 + D^2);
39      r2 = sqrt(X.^2 + (Y + d/2).^2 + D^2);
40
41      % Calculate path difference and Intensity
42      delta_r = abs(r1 - r2);
43      I = cos(pi * delta_r / lambda).^2;
44
45      % Plot the static interference pattern
46      subplot(2, 4, i);
47      imagesc([-screen_max screen_max], [-screen_max screen_max], I);
48      colormap('gray');
49
50      % Formatting
51      title(sprintf('Set %d:\n\\lambda=%.6f mm\nd=%.2f mm, D=%d mm', i
          , lambda, d, D));
52      xlabel('Screen X (mm)');
53      ylabel('Screen Y (mm)');
54      axis image;
55  end
56
57  % Reveal the fully drawn figure all at once for a smooth pop-up
58  set(fig, 'Visible', 'on');
59  % --- NEW CODE TO SAVE THE FIGURE ---
60  filename = 'interference_patterns_static.png';
61  print(fig, filename, '-dpng', '-r300'); % Saves as PNG at 300 DPI
62  fprintf('Saved high-resolution plot to: %s\n', fullfile(pwd,
          filename));
63  pause;

```

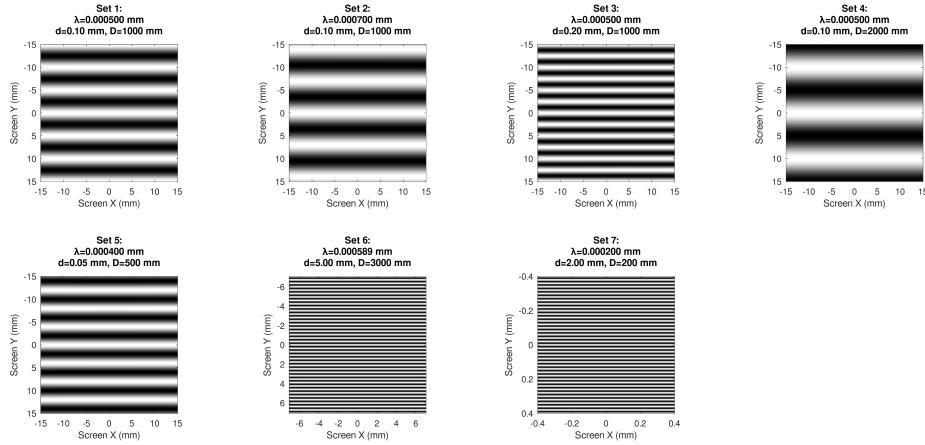


Figure 1: Output: greyscale interference fringe patterns for all 7 parameter sets arranged in a 2×4 subplot grid. White bands = constructive interference (bright fringes); black bands = destructive interference (dark fringes). Sets 1–5 show near-parallel fringes (Fraunhofer regime); Sets 6 and 7 show curved hyperbolic fringes (near-field / Fresnel effects visible at the screen edges due to the large slit separation relative to λD).

Sample Output

Result

The program was executed successfully and fringe patterns were obtained for all seven parameter sets. The following observations confirm the theoretical relations:

- **Set 1 (Baseline, $\beta = 5.00$ mm):** Clear, evenly spaced, symmetric bright and dark fringes.
- **Set 2 ($\uparrow \lambda \Rightarrow \beta = 7.00$ mm):** Wider fringes, confirming $\beta \propto \lambda$.
- **Set 3 ($\uparrow d \Rightarrow \beta = 2.50$ mm):** Narrower, denser fringes, confirming $\beta \propto 1/d$.
- **Set 4 ($\uparrow D \Rightarrow \beta = 10.0$ mm):** Wider fringes, confirming $\beta \propto D$.
- **Set 5 (Custom, $\beta = 4.00$ mm):** Intermediate spacing.
- **Set 6 (Na D-line, $\beta = 0.354$ mm):** Very dense fringes; dynamic zoom sets screen to ± 0.71 mm. Fringe curvature (hyperbolic shape) is noticeable at screen edges, indicating breakdown of the Fraunhofer far-field approximation.
- **Set 7 (UV, $\beta = 0.020$ mm):** Extremely dense fringes; near-field (Fresnel) effects dominate. Hyperbolic structure is clearly visible.

The output figure was also saved as `interference_patterns_static.png` at 300 DPI for inclusion in reports.

Remark

The program uses the *exact* spherical wave distances r_1 and r_2 (including the D^2 term under the square root) rather than the far-field Fraunhofer approximation $\Delta r \approx dY/D$. This distinction becomes significant in Sets 6 and 7, where the fringe curvature (hyperbolic shape) departs from straight parallel lines. The dynamic screen zoom is a critical numerical engineering decision: without it, sets with small $\lambda D/d$ would produce Moiré aliasing artefacts rather than physical fringes. The experiment confirms the theoretical fringe-width relation $\beta = \lambda D/d$ and illustrates the role of all three parameters independently.
